

**Owner-approved ExamSIDE question-demand
slice for plane motion, now semantically:**

Transfer

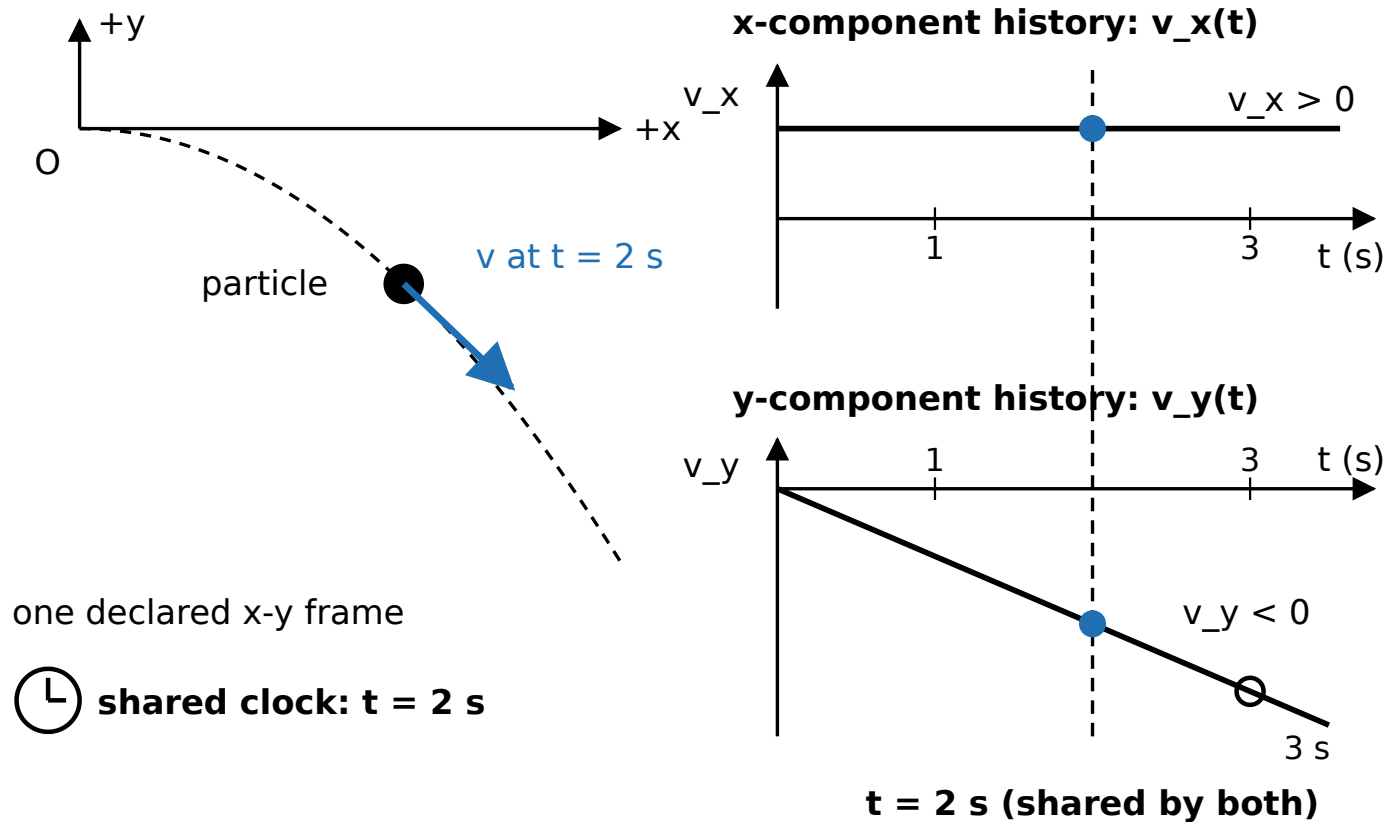
A particle is described verbally: at $t=0$ it is at the origin, moving east at 6 m/s and north at 2 m/s; during the next interval its acceleration is 3 m/s^2 south and zero east-west. Without being given component equations, choose a coordinate convention and write the signed component state that should be used for later calculation.

Builds on

- [A tracker reports \$x\(t\)=4t\$ metres and \$y\(t\)=10-2t^2\$ metres for one particle in one fixed...](#)

Conditions

- Use one fixed Cartesian frame.
- The verbal directions must be translated into signed components before calculation.



✓ $(v_x(2\text{ s}), v_y(2\text{ s}))$: simultaneous, one valid plane state

✗ $(v_x(2\text{ s}), v_y(3\text{ s}))$: non-simultaneous, rejected (no single instant)

Keep independently evolving x and y descriptions attached to one physical object, one declared frame and one shared clock, so only simultaneous component values are recombined into a plane state.

Commit to a model or representation and your first relation before opening support.

Review and solution

What changed

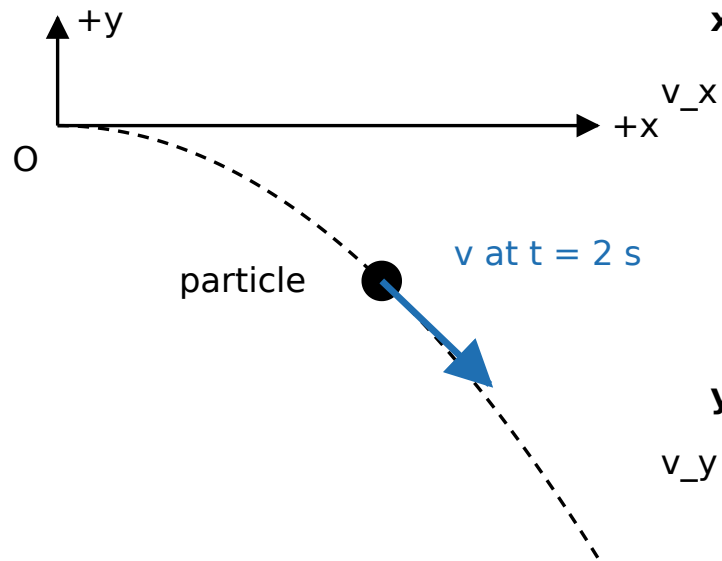
The learner must convert a verbal direction description into signed x/y state without being handed the component representation.

What stayed valid

One fixed frame and one shared clock still govern both components; only the source of the signed component values changes, from given functions to a verbal description.

The deciding move

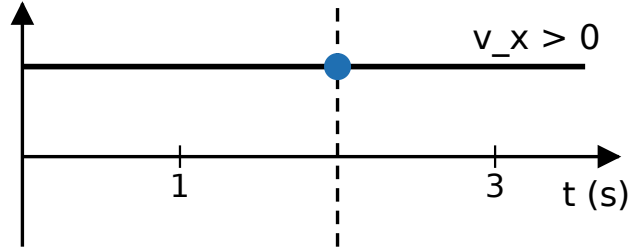
Choose and declare one perpendicular positive-axis convention before converting the directional facts into signed components.



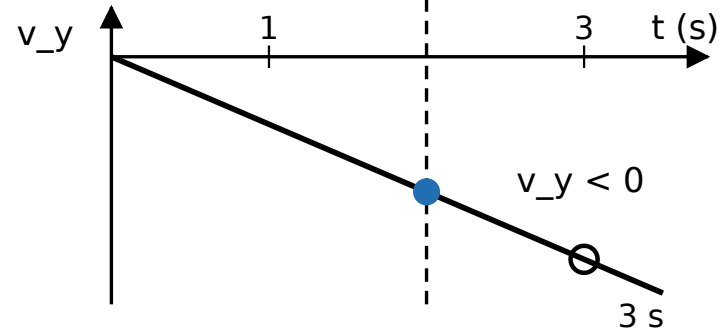
one declared x-y frame

⌚ shared clock: $t = 2$ s

x-component history: $v_x(t)$



y-component history: $v_y(t)$



$t = 2$ s (shared by both)

✓ $(v_x(2 \text{ s}), v_y(2 \text{ s}))$: simultaneous, one valid plane state

✗ $(v_x(2 \text{ s}), v_y(3 \text{ s}))$: non-simultaneous, rejected (no single instant)

Keep independently evolving x and y descriptions attached to one physical object, one declared frame and one shared clock, so only simultaneous component values are recombined into a plane state.

Answer

One valid choice is +x east and +y north, giving $u_x=+6$ m/s, $u_y=+2$ m/s, $a_x=0$ and $a_y=-3$ m/s², all with one common t.

1. Declare +x east and +y north.
2. Translate each direction into a sign in that frame.
3. Keep the same frame and clock for both component histories.

Rubric

- Declares a single Cartesian frame before assigning signs.
- Translates east/north/south statements into consistent signed components.
- Keeps one common time for the two component histories.

Independent check

Any consistently declared perpendicular axis convention is acceptable if all signs are transformed consistently.

Repair

Revisit: Represent position, velocity and acceleration as signed x and y components in that same frame.

Lineage check

One fixed frame and one shared clock still govern both components; only the source of the signed component values changes, from given functions to a verbal description.

The learner must convert a verbal direction description into signed x/y state without being handed the component representation.

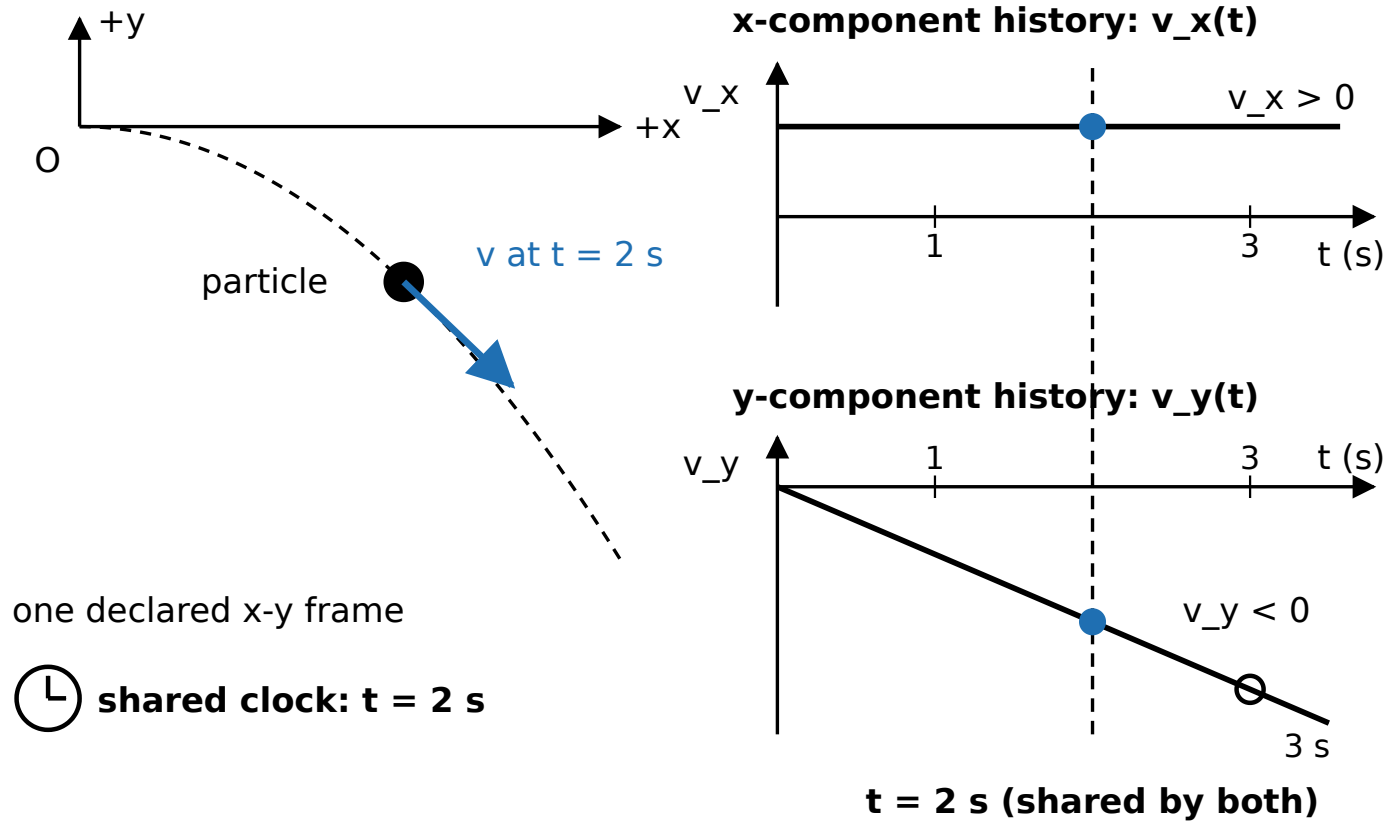
Over $0 \leq t \leq 4$ s, a particle has $a_x = 2 \text{ m/s}^2$ but $a_y = 3t \text{ m/s}^2$. Decide whether the ordinary constant-acceleration equations may be used exactly on x , on y , on both, or on neither over the whole interval. Explain the model choice before doing any further calculation.

Builds on

- [A particle starts with velocity \$\(3 \mathbf{i} + 4 \mathbf{j}\) \text{ m/s}\$ and has constant acceleration \$\(2 \mathbf{i} - 1 \mathbf{j}\) \text{ m/s}^2\$](#)

Conditions

- The acceleration components are given as functions of time.
- No numerical integration is required; the task is model selection.



✓ $(v_x(2\text{ s}), v_y(2\text{ s}))$: simultaneous, one valid plane state

✗ $(v_x(2\text{ s}), v_y(3\text{ s}))$: non-simultaneous, rejected (no single instant)

Keep independently evolving x and y descriptions attached to one physical object, one declared frame and one shared clock, so only simultaneous component values are recombined into a plane state.

Commit to a model or representation and your first relation before opening support.

Review and solution

What changed

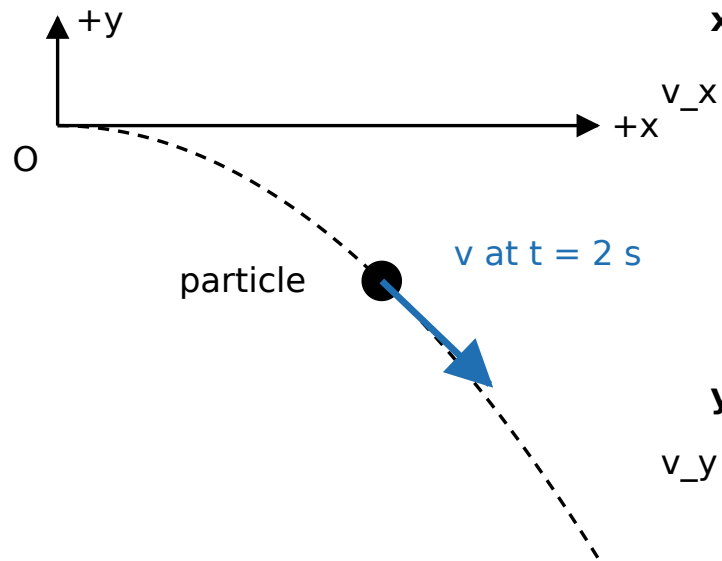
The learner must decide component-by-component whether the constant-acceleration model is valid instead of being told that it applies.

What stayed valid

Each axis is still handled with its own signed values over one shared time interval; what changes is that the constant-acceleration condition must be checked rather than assumed.

The deciding move

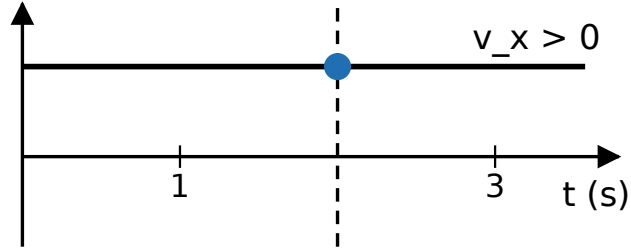
Decide separately for each axis whether the constant-acceleration condition is satisfied over the whole interval.



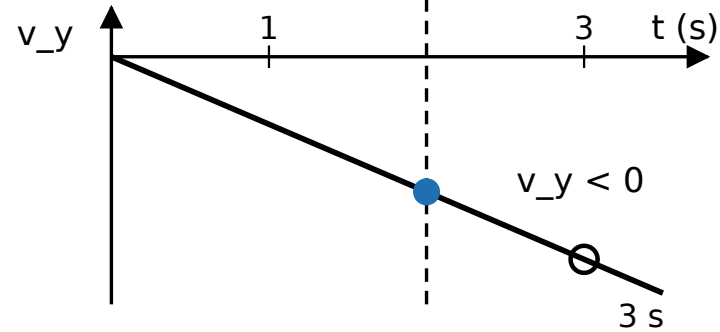
one declared x-y frame

⌚ shared clock: $t = 2$ s

x-component history: $v_x(t)$



y-component history: $v_y(t)$



$t = 2$ s (shared by both)

✓ $(v_x(2 \text{ s}), v_y(2 \text{ s}))$: simultaneous, one valid plane state

✗ $(v_x(2 \text{ s}), v_y(3 \text{ s}))$: non-simultaneous, rejected (no single instant)

Keep independently evolving x and y descriptions attached to one physical object, one declared frame and one shared clock, so only simultaneous component values are recombined into a plane state.

Answer

The constant-acceleration equations are exact on x because a_x is constant, but not on y because a_y changes with time over the interval.

1. Check each acceleration component separately.
2. $a_x=2$ is constant, so the 1D constant-acceleration x relations are valid.
3. $a_y=3t$ varies with time, so the constant- a y relations are not exact over the whole interval.

Rubric

- Checks x and y acceleration components separately.
- Accepts constant-acceleration equations only on x .

Independent check

A constant total direction or a constant x component does not make the y component constant.

Repair

[Revisit: Check separately whether \$a_x\$ and \$a_y\$ are constant over the interval before choosing constant-acceleration equations on either axis.](#)

Lineage check

Each axis is still handled with its own signed values over one shared time interval; what changes is that the constant-acceleration condition must be checked rather than assumed.

The learner must decide component-by-component whether the constant-acceleration model is valid instead of being told that it applies.

An ideal projectile begins 15 m above level ground with $u_x=10$ m/s and $u_y=10$ m/s. Take +y upward and $g=10$ m/s². Find the ground-impact time and horizontal range. Do not assume launch and landing are at the same height.

Builds on

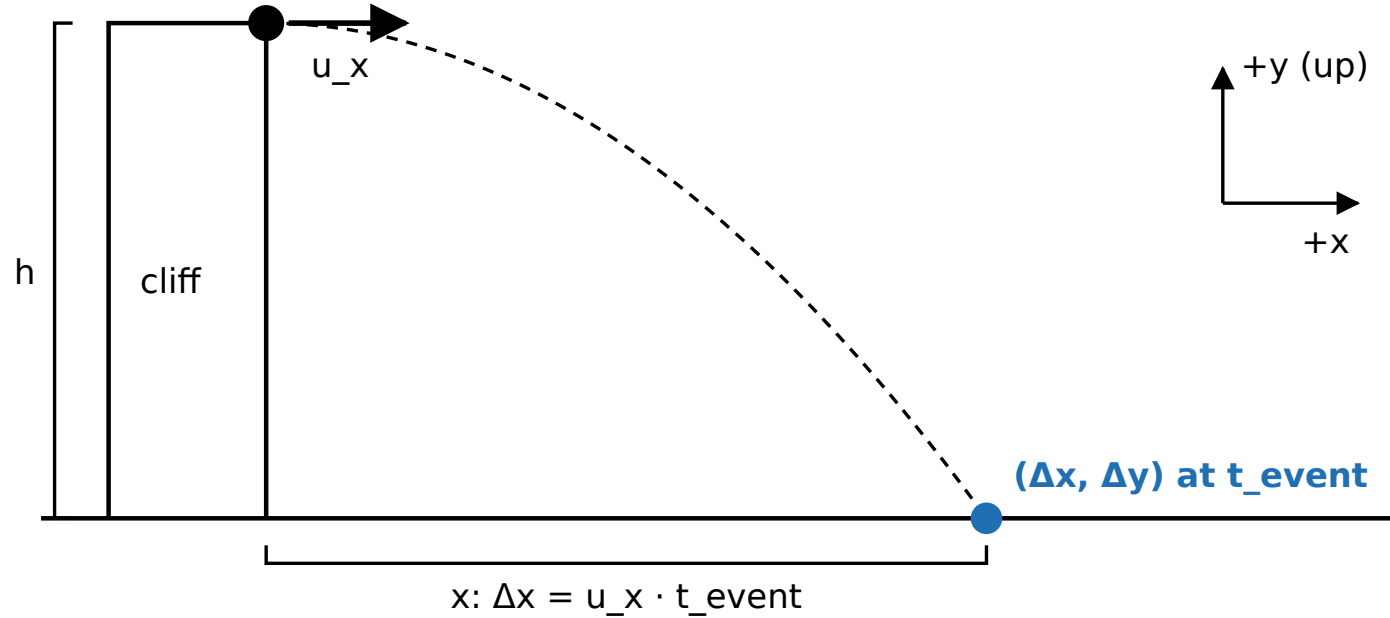
- [An ideal projectile is launched and later lands at the same height with \$u_x=12\$ m/s,...](#)

Conditions

- Air resistance is neglected.
- Ground is 15 m below the launch point.
- The initial velocity components are already resolved.

Requested event: the stone reaches the ground

event condition (y): $\Delta y = -h$



one shared event timeline

event time t_{event}



y: $\Delta y = -h$ at the same t_{event}

The one t_{event} is reused unchanged on x and on y.

Make event selection explicit: one component condition identifies the requested event time, and that same time is then reused on every component before reconstructing the requested state.

Commit to a model or representation and your first relation before opening support.

Review and solution

What changed

The learner must replace the familiar same-height condition with an unequal-height event condition and carry the resulting time across axes.

What stayed valid

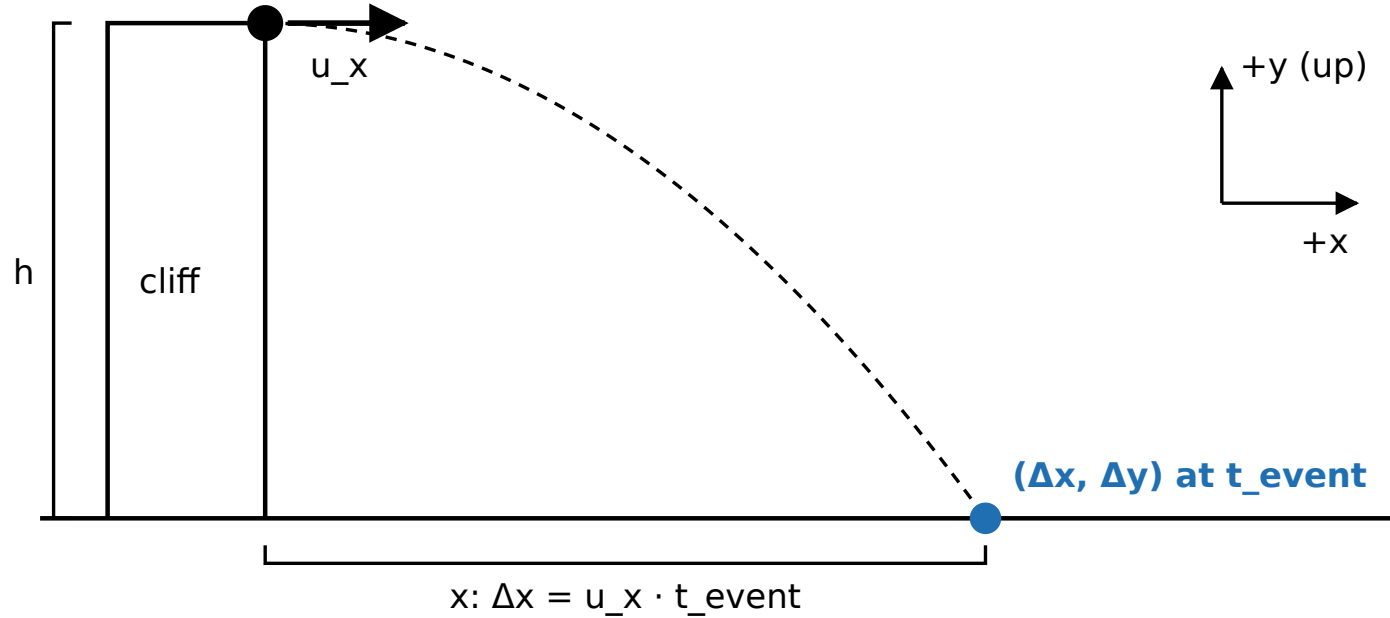
The gravity-only model, $a_x = 0$ and $a_y = -g$, and the rule that one event time serves both axes still hold; only the vertical condition that defines landing changes.

The deciding move

Choose $\Delta y = -15 \text{ m}$ as the impact condition instead of reusing the familiar same-height condition $\Delta y = 0$.

Requested event: the stone reaches the ground

event condition (y): $\Delta y = -h$



one shared event timeline

event time t_{event}



$y: \Delta y = -h$ at the same t_{event}

The one t_{event} is reused unchanged on x and on y.

Stage 1 of 4

Make event selection explicit: one component condition identifies the requested event time, and that same time is then reused on every component before reconstructing the requested state.

Answer

Ground impact occurs at $t=3$ s and the horizontal range is 30 m.

1. Use the vertical impact condition $\Delta y = -15 = 10t - 5t^2$.
2. Solve $5t^2 - 10t - 15 = 0$, giving the physical root $t=3$ s.
3. Reuse $t=3$ s horizontally: $\Delta x = 10(3) = 30$ m.
4. The same-height shortcut would use the wrong event condition here.

Rubric

- Uses the actual vertical displacement -15 m rather than a same-height shortcut.
- Reuses the physical impact time in the horizontal equation.

Independent check

Substitution into the vertical displacement gives -15 m; the nonzero same-height time $2u_y/g = 2$ s is not the ground-impact time.

Repair

Revisit: For any unequal-height landing, use the actual $\Delta y = y_f - y_i$ in the vertical position equation and retain the actual u_y ; for a horizontal launch specifically, set $u_y = 0$, then obtain the event time vertically and share it with x .

Lineage check

The gravity-only model, $a_x = 0$ and $a_y = -g$, and the rule that one event time serves both axes still hold; only the vertical condition that defines landing changes.

The learner must replace the familiar same-height condition with an unequal-height event condition and carry the resulting time across axes.

An object is launched and, after release, a small rocket motor continues to provide a horizontal thrust that gives $a_x=2 \text{ m/s}^2$ while gravity gives $a_y=-g$. Decide whether the standard ideal projectile specialization $a_x=0$, $a_y=-g$ is valid and state the correct bounded model to use.

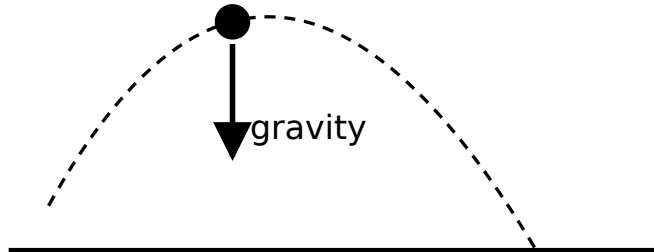
Builds on

- [A stone is launched horizontally at 15 m/s from a cliff 20 m above level ground. Take...](#)

Conditions

- Air resistance is neglected.
- The horizontal rocket thrust remains active after release.
- The question asks for model selection, not a full trajectory calculation.

A: launched, then free flight



After release, acting on the object:

- gravity (Earth on object), down
- drag: neglected

Gravity only? yes

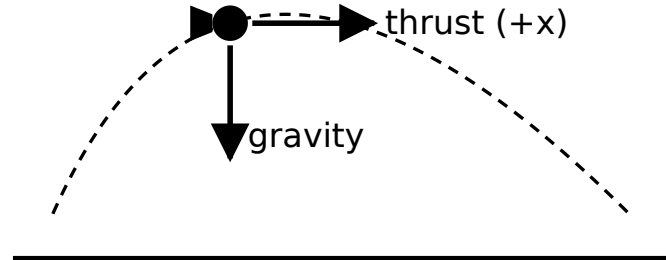
$$a_x = 0$$

$$a_y = -g \text{ (+y up)}$$

Ideal projectile model: VALID

$a_x = 0$ and $a_y = -g$ both hold after release.

B: launched, motor stays on



After release, acting on the object:

- gravity (Earth on object), down
- thrust (motor on object), horizontal
- drag: neglected

Gravity only? no (thrust retained)

$$a_x \neq 0 \text{ (from the thrust)}$$

$$a_y = -g \text{ (+y up)}$$

Ideal projectile model: REJECTED

Active thrust violates $a_x = 0$.
Use the general 2D
constant-acceleration model.

Make the ideal projectile specialization conditional on the interactions that remain after release: gravity-only free flight licenses $a_x=0$ and $a_y=-g$; retained thrust, significant drag or contact requires a more general model.

Commit to a model or representation and your first relation before opening support.

Review and solution

What changed

The learner must reject the familiar projectile specialization because one of its defining force/model conditions is explicitly broken.

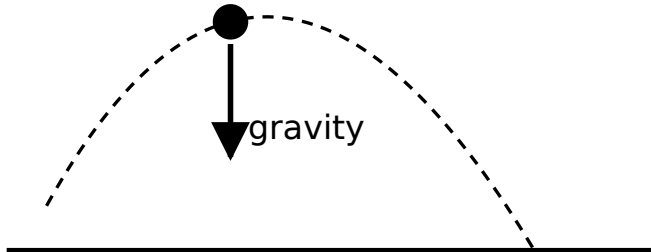
What stayed valid

Component-by-component constant-acceleration kinematics with one shared clock still applies; what fails is only the special case $a_x = 0$ that the projectile model adds.

The deciding move

Reject the standard gravity-only projectile specialization because its defining horizontal-acceleration condition is violated.

A: launched, then free flight



After release, acting on the object:

- gravity (Earth on object), down
- drag: neglected

Gravity only? yes

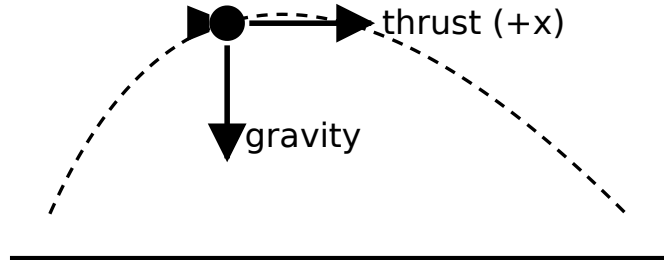
$$a_x = 0$$

$$a_y = -g \text{ (+y up)}$$

Ideal projectile model: **VALID**

$a_x = 0$ and $a_y = -g$ both hold after release.

B: launched, motor stays on



After release, acting on the object:

- gravity (Earth on object), down
- thrust (motor on object), horizontal
- drag: neglected

Gravity only? no (thrust retained)

$$a_x \neq 0 \text{ (from the thrust)}$$

$$a_y = -g \text{ (+y up)}$$

Ideal projectile model: **REJECTED**

Active thrust violates $a_x = 0$.
Use the general 2D
constant-acceleration model.

Make the ideal projectile specialization conditional on the interactions that remain after release: gravity-only free flight licenses $a_x=0$ and $a_y=-g$; retained thrust, significant drag or contact requires a more general model.

Answer

The standard ideal projectile specialization is not valid because a_x is not zero. Use the more general 2D constant-acceleration model with $a_x=2 \text{ m/s}^2$ and $a_y=-g$ for as long as those components remain constant.

1. The standard projectile specialization requires gravity-only free flight.
2. The active rocket motor adds a retained horizontal acceleration.
3. Therefore use the parent 2D constant-acceleration capability rather than the gravity-only projectile specialization.

Rubric

- Rejects the gravity-only projectile specialization.
- Falls back to the general component constant-acceleration model.

Independent check

If the motor is switched off after release and drag remains negligible, the standard $a_x=0$ projectile model becomes valid from that later instant.

Repair

Revisit: Decide whether the object is in ideal near-Earth free flight after release; neglect air resistance, declare $+y$, then set $a_x=0$ and $a_y=-g$.

Lineage check

Component-by-component constant-acceleration kinematics with one shared clock still applies; what fails is only the special case $a_x = 0$ that the projectile model adds.

The learner must reject the familiar projectile specialization because one of its defining force/model conditions is explicitly broken.

An ideal projectile passes the same height once on the way up and once on the way down. Without calculating the times, compare v_x , v_y , speed and acceleration at those two events.

Builds on

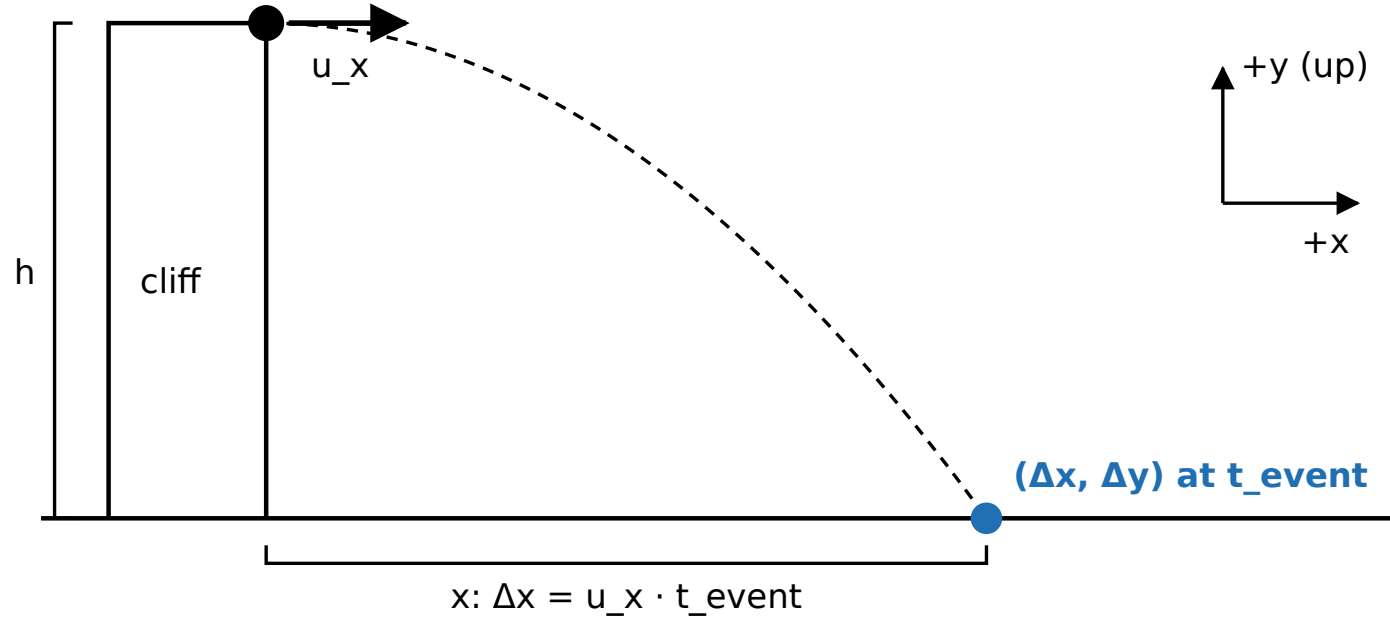
- [An ideal projectile is launched with already-resolved components \$u_x=12\$ m/s and \$u_y=20\$...](#)

Conditions

- Air resistance is neglected.
- Both events occur on the same trajectory at the same vertical position.
- Take +y upward.

Requested event: the stone reaches the ground

event condition (y): $\Delta y = -h$



one shared event timeline

event time t_{event}



y: $\Delta y = -h$ at the same t_{event}

The one t_{event} is reused unchanged on x and on y.

Make event selection explicit: one component condition identifies the requested event time, and that same time is then reused on every component before reconstructing the requested state.

Commit to a model or representation and your first relation before opening support.

Review and solution

What changed

The learner must compare two equal-height events by component invariants rather than solve a routine time/range calculation.

What stayed valid

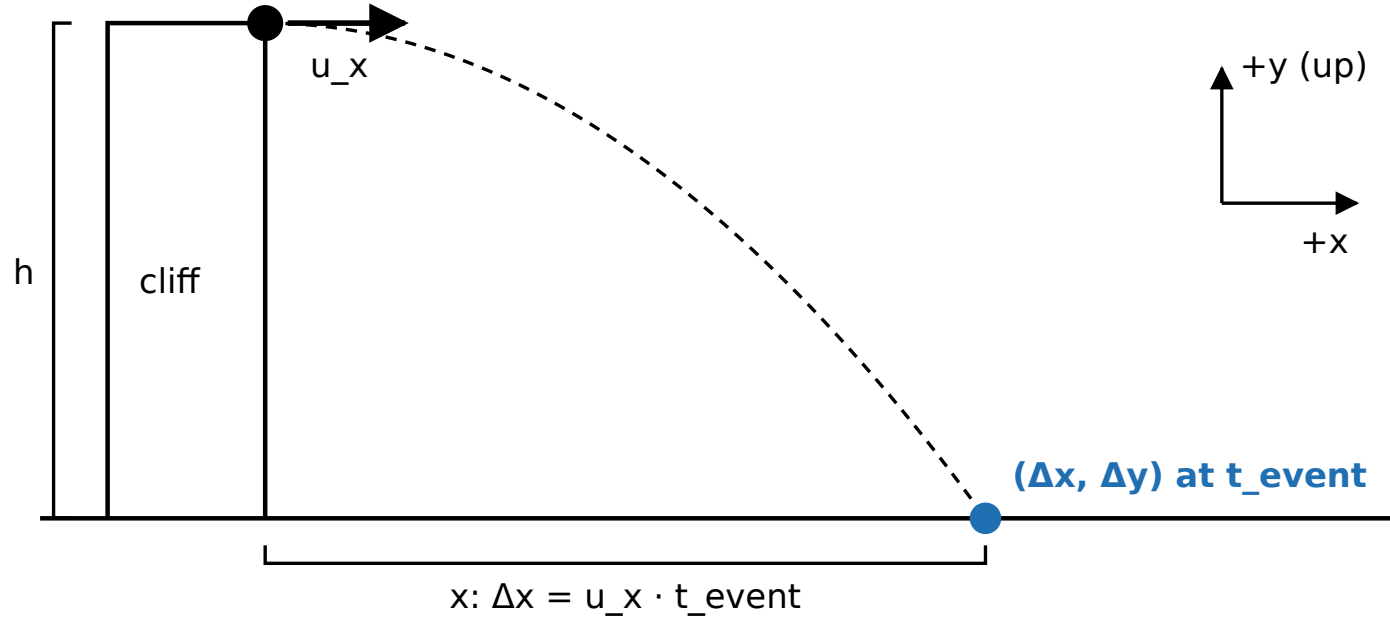
v_x stays constant and $a = (0, -g)$ at every instant of ideal flight, exactly as at the apex; the task changes from locating one event to comparing two.

The deciding move

Choose direct component invariants at equal height as the comparison method instead of solving for the two event times.

Requested event: the stone reaches the ground

event condition (y): $\Delta y = -h$



one shared event timeline

event time t_{event}



y: $\Delta y = -h$ at the same t_{event}

The one t_{event} is reused unchanged on x and on y.

Stage 1 of 4

Make event selection explicit: one component condition identifies the requested event time, and that same time is then reused on every component before reconstructing the requested state.

Answer

v_x is the same at both events; v_y has equal magnitude and opposite sign; therefore the speed is the same but the velocity vectors differ. Acceleration is the same (0,-g) at both events.

1. Horizontal velocity remains constant because $a_x=0$.
2. At equal height in the same gravity-only trajectory, the vertical speed magnitude matches on ascent and descent, with opposite signs.
3. Combining the components gives equal speed magnitudes but different velocity directions.
4. Gravity gives the same downward acceleration at both events.

Rubric

- Keeps v_x unchanged across the two events.
- Gives v_y equal magnitude and opposite sign.
- Distinguishes equal speed from equal velocity.

Independent check

The two velocity vectors cannot be identical because one has upward v_y and the other downward v_y .

Repair

Revisit: At the selected event time, reconstruct the requested velocity, speed, direction, displacement or range from simultaneous components and check qualitative limits.

Lineage check

v_x stays constant and $a = (0, -g)$ at every instant of ideal flight, exactly as at the apex; the task changes from locating one event to comparing two.

The learner must compare two equal-height events by component invariants rather than solve a routine time/range calculation.